



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO HARUR APCMS HARUR

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

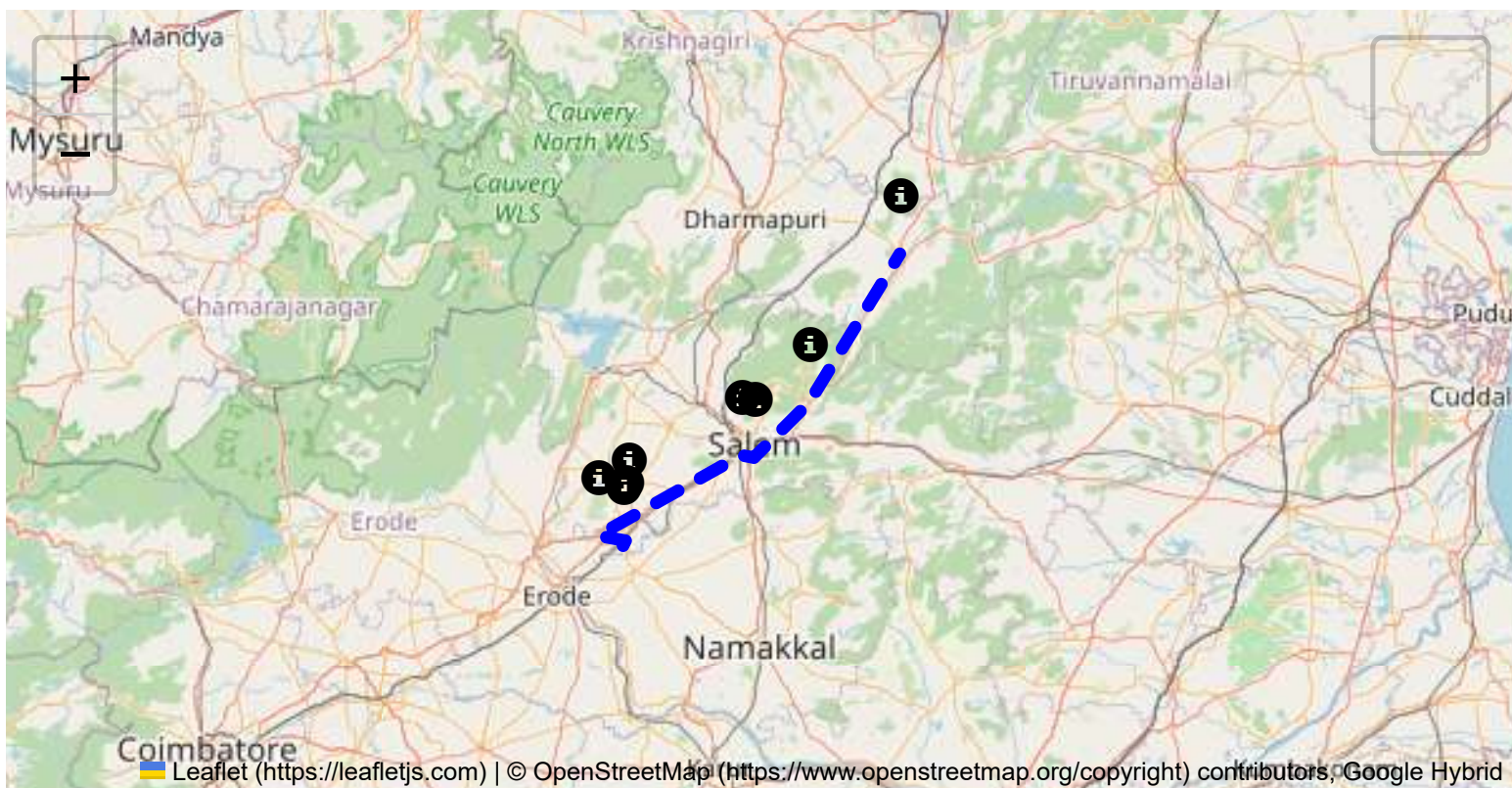
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

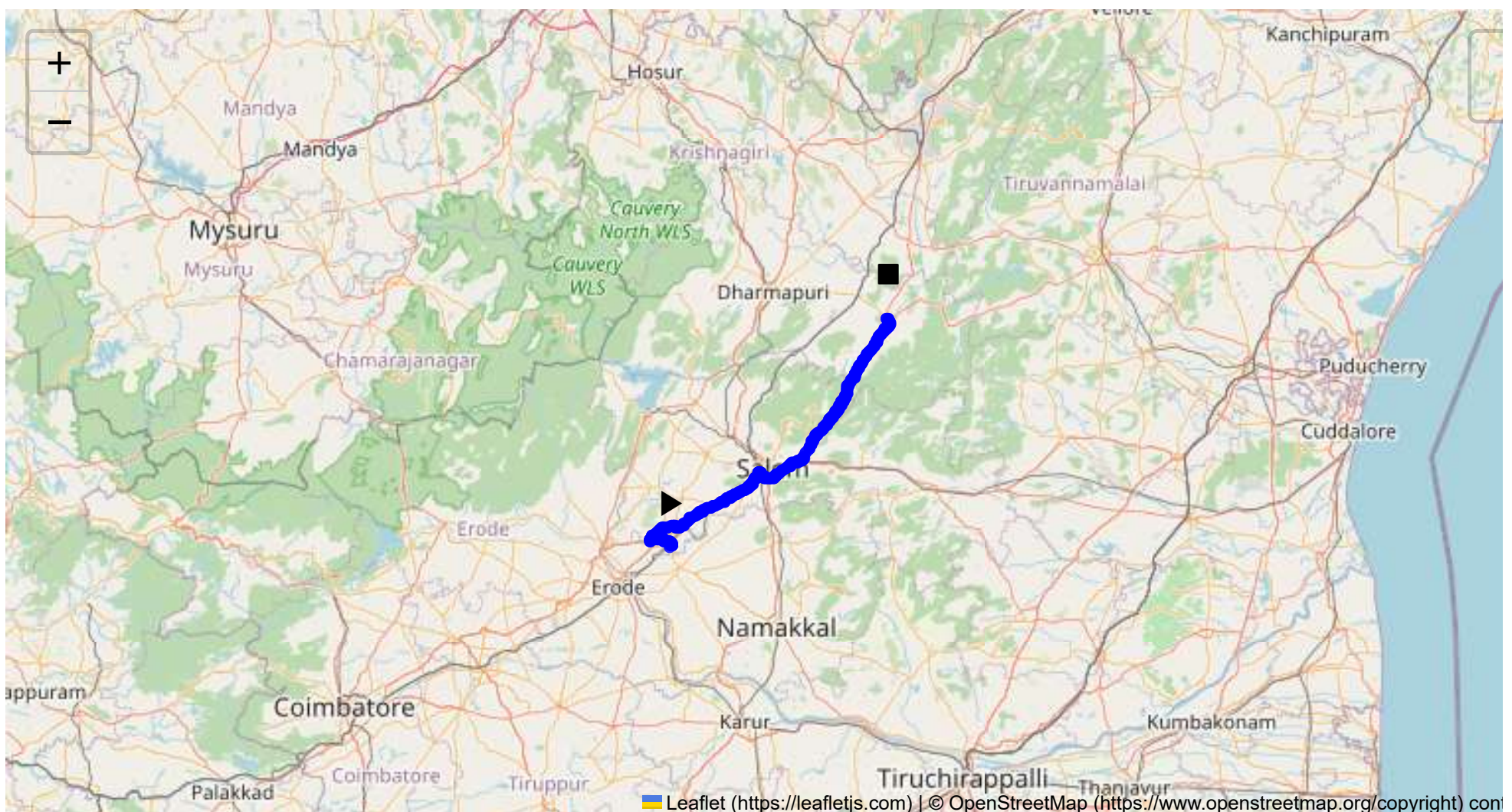
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 118.02 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.05838, 78.477815)



Welcome to the Journey Risk Management Study

Sure, here is a detailed analysis of the specified route:

1. Overview of the Route Map

The route covers approximately 118.02 kilometers from Sangagiri to T V K Nagar, passing through several significant waypoints in Tamil Nadu, India. The journey traverses a mix of state highways and local roads,

including stretches on Bhavani Main Road and Salem-Kochi Highway. These roads connect various small towns and suburbs with the major city of Salem, which serves as a critical junction.

2. Typical Weather Conditions and Hazards

Tamil Nadu experiences a tropical climate, with the potential for heavy rainfall during the monsoon season (from October to December). This can lead to slick roads, water-logging, and reduced visibility. It's advisable to monitor weather forecasts, especially during these months, to avoid heavy showers and related road conditions.

3. Traffic Patterns and Congestion

- **Peak Hours:** Traffic congestion is notable during morning (8 AM - 10 AM) and evening (5 PM - 7 PM) rush hours, especially in and around Salem city.
- **Congestion-Prone Areas:** Key areas include the Salem city bypass and entry/exit points of major highways (Salem-Kochi Highway and NH44). Routes entering busy market areas can also experience slowdowns.

4. Road Quality and Infrastructure

The road infrastructure varies:

- **Highways:** Generally well-maintained with clear signage and lane markings. However, be aware of ongoing roadworks which might occur in patches.
- **Local Roads:** Some sections might have potholes and uneven surfaces. These require caution, especially for heavy trucks.

5. Alternative Routes for Emergencies

- **Option 1:** For avoiding Salem during high congestion, consider taking the rural roads parallel to the main highway. However, these may not support large vehicles well, so use with discretion.
- **Option 2:** Utilize the NH44 for any detours, as it offers multiple connections to smaller towns that can be used to bypass traffic jams or other issues.

6. Local Regulations on Hazardous Material Transport

Tamil Nadu has stringent checks for hazardous material transport. Ensure compliance with permits related to cargo type, and note that there are restrictions during high-traffic hours within the city limits.

7. Historical Incidents

Salem and its surrounding areas have experienced occasional incidents involving heavy vehicles, primarily due to speeding and mechanical failures. Adjust speeds accordingly and ensure vehicle maintenance to mitigate these risks.

8. Environmental Considerations

The route crosses agricultural zones and densely populated areas. Be cautious of noise pollution and adhere to speed limits, especially near villages and towns to minimize environmental impact. Watch for stray animals, especially in rural stretches.

9. Communication Coverage

Most of the route provides stable mobile network coverage, although there may be patchy service in isolated rural areas. Carrying a GPS device and an offline map is recommended for navigation support in dead zones.

10. Emergency Response Times

- Urban Areas (e.g., Salem):** Typically 10-15 minutes.
- Rural Sections:** Response times could range from 25 to 45 minutes due to distances from emergency facilities.

11. Overall Summary of Risk Assessment

This route is generally manageable for trucks carrying hazardous materials, provided they follow regulatory guidelines and drive cautiously. The combination of well-maintained highways and more rugged local roads means that careful navigation is essential, particularly during inclement weather and peak traffic hours. Proactive planning around high-risk zones and awareness of environmental and traffic conditions will ensure safer transit along this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.63303, 78.12713	15 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	High	11.44898, 77.87410	15 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	High	11.46303, 77.84945	15 KM/Hr
8	Turn	Medium	11.45543, 77.81740	30 KM/Hr
9	Turn	High	11.45631, 77.81668	15 KM/Hr
10	Turn	Medium	11.53569, 77.95787	30 KM/Hr
11	Turn	Medium	11.64172, 78.11817	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	Medium	11.64216, 78.11875	30 KM/Hr
13	Turn	Medium	11.64195, 78.12129	30 KM/Hr
14	Turn	High	11.64162, 78.12160	15 KM/Hr
15	Turn	Medium	11.64087, 78.12157	30 KM/Hr
16	Turn	High	11.63948, 78.12081	15 KM/Hr
17	Turn	Medium	11.62821, 78.13348	30 KM/Hr
18	Turn	Medium	11.62836, 78.16483	30 KM/Hr
19	Turn	High	11.66380, 78.20648	15 KM/Hr
20	Turn	High	11.67120, 78.23136	15 KM/Hr
21	Turn	High	11.67190, 78.23144	15 KM/Hr
22	Turn	Medium	11.67429, 78.23896	30 KM/Hr
23	Turn	Medium	11.81208, 78.34071	30 KM/Hr
24	Turn	Medium	11.83518, 78.35305	30 KM/Hr
25	Turn	Medium	11.99561, 78.44554	30 KM/Hr
26	Turn	Medium	12.01522, 78.45634	30 KM/Hr
27	Turn	Medium	12.04298, 78.48358	30 KM/Hr
28	Turn	Medium	12.04936, 78.48347	30 KM/Hr
29	Turn	High	12.04964, 78.48309	15 KM/Hr
30	Blind Spot	Blind Spot	12.05995, 78.47789	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
4	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
5	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
8	hospital	Vinayaka Mission's Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	hospital	Priyam Speciality Hospital	11.63446, 78.124388	30 km/h	Medium
14	hospital	Universal Cancer Hospital	11.6336739, 78.1267767	30 km/h	Medium
15	hospital	AVM Hospital	11.6264369, 78.1394204	30 km/h	Medium
16	hospital	மருத்துவமனை நிலம்	11.625344, 78.146058	30 km/h	Medium
17	hospital	Dharan Hospital	11.6252302, 78.1451753	30 km/h	Medium
18	hospital	Udhayam Hospital	11.625819, 78.145958	30 km/h	Medium
19	hospital	Manima Clinic	11.623248, 78.1465721	30 km/h	Medium
20	pharmacy	மருந்தகம்	11.6251501, 78.1470098	30 km/h	Medium
21	hospital	Rps Hospital	11.6260528, 78.1474921	30 km/h	Medium
22	hospital	Akitha Hospital	11.6256027, 78.1489875	30 km/h	Medium
23	hospital	Deepam Speciality Hospital	11.6269734, 78.1476357	30 km/h	Medium
26	hospital	M. Thathanur Primary Health Center	11.6824017, 78.2445813	30 km/h	Medium
27	hospital	Goverment Hospital	11.7580777, 78.2969128	30 km/h	Medium
28	hospital	Government Hospital Chinnankuppam	12.0110294, 78.4553224	30 km/h	Medium
35	hospital	Nanjaya Hospital	12.0550888, 78.4821929	30 km/h	Medium
36	hospital	Subam Hospital	12.0595128, 78.4766713	30 km/h	Medium
37	hospital	Government Hospital, Harur	12.061178, 78.4756913	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
6	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
9	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
10	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
11	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
13	school	Sowdeshwari College	11.6330643, 78.1276021	30 km/h	Medium
24	school	Ganesh College	11.6566575, 78.1991121	30 km/h	Medium
25	school	MettupattiThathanur Primary School	11.6827707, 78.2442509	30 km/h	Medium
29	school	Indian Matriculation Higher Secondary school	12.0061755, 78.4533388	30 km/h	Medium
30	school	Kurinji Matriculation Higher Secondary School	12.0156853, 78.4579056	30 km/h	Medium
31	school	Chinnan Kuppam Government High School	12.0190729, 78.4581955	30 km/h	Medium
32	college	Annai Arts and Science College	12.0217961, 78.462795	30 km/h	Medium
33	school	Jayam Vidhyalaya Matriculation Higher Secondary School	12.0264103, 78.4656553	30 km/h	Medium
34	school	Harur International School (CBSE)	12.0299967, 78.4693492	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.63303, 78.12713



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.53569, 77.95787



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64172, 78.11817



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64216, 78.11875



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64195, 78.12129



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64162, 78.12160



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64087, 78.12157



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.63948, 78.12081



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.62821, 78.13348



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62836, 78.16483



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66380, 78.20648



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.67120, 78.23136



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.67190, 78.23144



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67429, 78.23896



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.81208, 78.34071



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.83518, 78.35305



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.99561, 78.44554



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.01522, 78.45634



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.04298, 78.48358



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.04936, 78.48347



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.04964, 78.48309



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.05995, 78.47789
